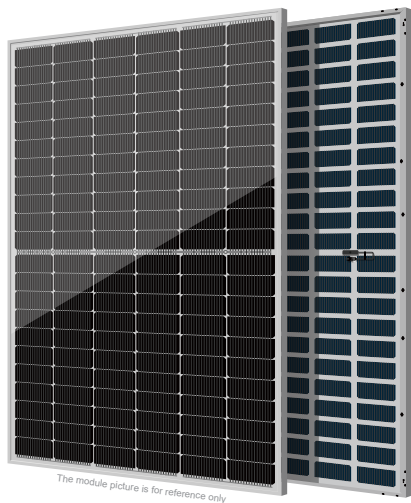




ZXNR-BD132 Series

SMBB HALF-CELL N-Type Bifacial Double Glass
Monocrystalline PV Module

620-650W

POWER RANGE

24.1%

MAXIMUM EFFICIENCY

0.40%

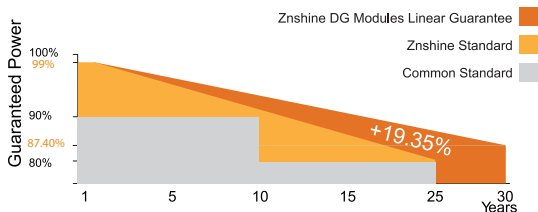
YEARLY DEGRADATION



12 YEARS PRODUCT WARRANTY



30 YEARS OUTPUT GUARANTEE



*Please check the Limited Warranty for Standard PV Modules which is officially released by ZNSHINE PV-TECH Co.,Ltd.



ISO 14001: Environmental Management System

ISO 9001: Quality Management System

ISO45001: Occupational Health and Safety Management System

*As there are different certification requirements in different markets, please contact your local znshine sales representative for the specific certificates applicable to the products in the region in which the products are to be used.

KEY FEATURES



Excellent Cells Efficiency

SMBB technology reduce the distance between busbars and finger grid line which is benefit to power increase.



Better Weak Illumination Response

More power output in weak light condition, such as haze, cloudy, and early morning.



Anti PID

Ensured PID resistance through the quality control of cell manufacturing process and raw materials.



Adapt To Harsh Outdoor Environment

Resistant to harsh environments such as salt, ammonia, sand, high temperature and high humidity environment.



TIER 1

Global, Tier 1 bankable brand, with independently certified advanced automated manufacturing.



Excellent Quality Management System

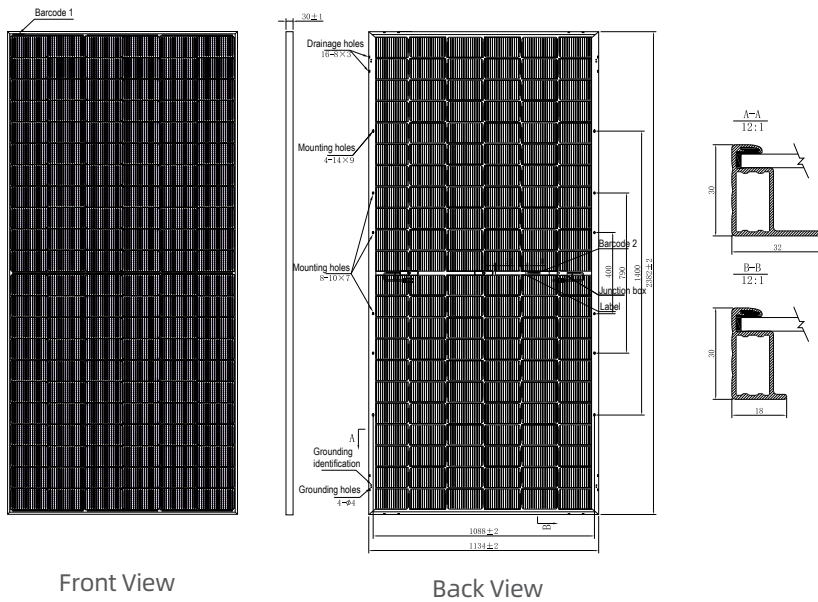
Warranted reliability and stringent quality assurances well beyond certified requirements.



Bifacial Technology

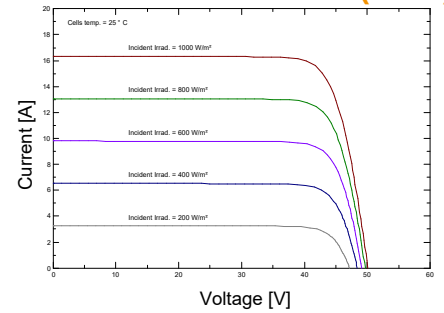
Up to 25% additional power gain from back side depending on albedo.

DIMENSIONS OF PV MODULE(mm)

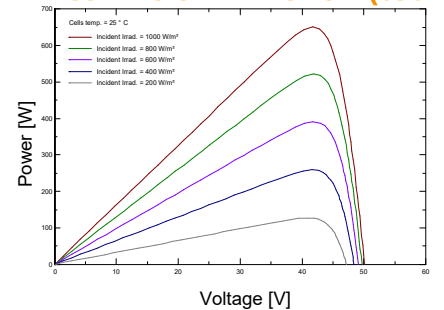


*Remark: customized frame color and cable length available upon request
only the classic 16BB assembly diagram is displayed here, other busbar can be customized according to requirements

I-V CURVES OF PV MODULE(650W)



P-V CURVES OF PV MODULE(650W)



ELECTRICAL CHARACTERISTICS | STC*

Nominal Power Watt Pmax(W)*	620	625	630	635	640	645	650
Maximum Power Voltage Vmp(V)	41.00	41.20	41.40	41.60	41.80	42.00	42.20
Maximum Power Current Imp(A)	15.13	15.17	15.22	15.27	15.32	15.36	15.41
Open Circuit Voltage Voc(V)	48.90	49.10	49.30	49.50	49.70	49.90	50.10
Short Circuit Current Isc(A)	16.05	16.09	16.14	16.19	16.24	16.29	16.34
Module Efficiency (%)	22.0	23.1	23.3	23.5	23.7	23.9	24.1

*The data above is for reference only and the actual data is in accordance with the practical testing

*STC (Standard Test Condition): Irradiance 1000W/m², Module Temperature 25±2°C, AM 1.5

*Measuring uncertainty: ±3%, all the electrical characteristics such as Power, Im, Vm, Isc, Voc and FF are within ±5% tolerance.

MECHANICAL DATA

Solar cells	N-type Monocrystalline, Rectangular cells
Cells orientation	132 (6×22)
Module dimension	2382×1134×30 mm (With Frame)
Weight	33.0±1.0 kg
Glass	2.0 mm+2.0mm, High Transmission, AR Coated Heat Strengthened Glass
Junction box	IP 68, 3 diodes
Cables	4 mm², 350 mm (With Connectors)
Connectors*	MC4-EVO2 compatible

*Please refer to regional datasheet for specified connector

ELECTRICAL CHARACTERISTICS | NMOT*

Maximum Power Pmax(Wp)	471.30	475.20	478.90	482.70	486.50	490.20	494.00
Maximum Power Voltage Vmpp(V)	38.04	38.50	38.70	38.90	39.10	39.30	39.50
Maximum Power Current Impp(A)	12.29	12.33	12.37	12.41	12.44	12.48	12.52
Open Circuit Voltage Voc(V)	46.30	46.50	46.70	46.90	47.10	47.30	47.40
Short Circuit Current Isc(A)	12.95	12.98	13.02	13.06	13.10	13.14	13.18

*NMOT:Irradiance 800W/m², Ambient Temperature 20°C, AM 1.5, Wind Speed 1 m/s

TEMPERATURE RATINGS

NMOT	44°C ±2°C	Maximum system voltage	1500 V DC
Temperature coefficient of Pmax	(-0.28±0.028)%/°C	Operating temperature	-40°C~+85°C
Temperature coefficient of Voc	-0.23%/°C	Maximum series fuse	30 A
Temperature coefficient of Isc	0.045%/°C	Front Side Maximum Static Loading	Up to 5400Pa
Refer.Bifacial Factor	(80±10)%	Rear Side Maximum Static Loading	Up to 2400Pa

*Remark: Do not connect Fuse in Combiner Box with two or more strings in parallel connection

WORKING CONDITIONS

ELECTRICAL CHARACTERISTICS (REAR POWER GAIN)

5%	Maximum Power:Pmax(W)	651	656	662	667	672	677	683
	Module Efficiency(%)	24.1	24.3	24.5	24.7	24.9	25.1	25.3
15%	Maximum Power:Pmax(W)	713	719	725	730	736	742	748
	Module Efficiency(%)	26.4	26.6	26.8	27.0	27.2	27.5	27.7
25%	Maximum Power:Pmax(W)	775	781	788	794	800	806	813
	Module Efficiency(%)	28.7	28.9	29.2	29.4	29.6	29.8	30.1

PACKAGING CONFIGURATION *

Piece/Box	36
Piece/Container(40'HQ)	720

*Customized packaging is available upon request.

*Remark:Electrical data in this catalog do not refer to a single module and they are not part of the offer.
They only serve for comparison among different module types.

*Caution:Please be kindly advised that PV modules should be handled and installed by qualified people who have professional skills and please carefully read the safety and installation instructions before using our PV modules.